

Magnetic Fields — Paper C

HARD • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Write in sentences — a list of single words scores badly on this topic.
- Every field line you draw must carry an **arrowhead**. A line without one does not score.
- Where a question asks for a number, that number must carry a **unit**.
- The mark for each question is shown in square brackets on the right.
- No rules and no reminders are given on this paper.
- Several questions describe experiments you may not have seen before — apply the ideas you already have.

SECTION 1 – READING THE EVIDENCE

- 1.** A steel bar hangs from a thread so that it can swing freely. When the **north** pole of a magnet is brought up to end X of the bar, the bar swings towards the magnet. When the **south** pole is brought up to the same end X, the bar swings towards the magnet again.

(a) State what these two observations show about the steel bar. [1]

(b) Explain the observations in terms of induced magnetism. [2]

(c) Describe how the observations would have been different if the bar had been a magnet. [2]

2. Three bars look identical. One is a magnet, one is unmagnetised steel and one is brass. Describe how you would identify all three, using nothing but the three bars themselves. State clearly what you would observe in each case. [4]

SECTION 2 – FIELDS, NEUTRAL POINTS AND COMPASSES

3. Two identical bar magnets lie in a line on a bench with their **north poles facing each other**, 6 cm apart.

(a) State and explain what happens when they are released. [2]

(b) A plotting compass placed at one particular point between them will not settle in any direction. State where that point is and explain why. [2]

(c) The magnets are pushed until the gap is 2 cm. Assuming the force follows an inverse-square law, state the factor by which the force increases. [1]

4. A bar magnet lies on a bench with its north pole on the left and its south pole on the right. A plotting compass is placed at three points in turn. State the direction the compass needle's **north end** points in each case.

(a) On the axis, just beyond the north pole. [1]

(b) On the axis, just beyond the south pole. [1]

(c) Directly above the middle of the magnet. [1]

(d) Explain how you worked out your answer to (c). [1]

SECTION 3 – APPLYING THE IDEAS

5. A crane at a scrapyards lifts material out of a mixed heap using a very strong magnet. The heap contains steel cans, aluminium cans, copper wire and iron girders.

(a) State which items the magnet lifts and which it leaves behind, with a reason. [2]

(b) The operator wants to separate the copper wire from the aluminium cans. Explain why a second pass with the magnet cannot do it. [2]

6. A magnet holds a hanging chain of three steel paper clips.

(a) Explain why each clip in the chain is a weaker magnet than the one above it. [2]

- (b)** A thin sheet of copper is slid between the magnet and the first clip. Predict what happens to the chain, and explain your prediction. [2]

- 7.** The Earth's magnetic field is usually drawn as though a huge bar magnet were buried along its axis.

- (a)** State which end of that buried magnet lies at the geographic North Pole, and justify it. [2]

- (b)** Give one way in which the buried-bar-magnet picture is useful, and one way in which it is not literally true. [2]

End of paper. Before you hand it in, go back over every diagram and check that each field line carries an arrowhead pointing from north to south.